

Text Mining Project: Predicting Song Popularity through Lyrics and Features

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Abstract

This document supplements our text mining project instructions. It outlines the use of song lyrics and features to predict views using the Genius Song Lyrics dataset. The dataset includes lyrics, titles, features, and artists, with views as the target variable. We apply preprocessing techniques such as lemmatization and feature cleaning, and use methods like TF-IDF, Bag-of-Words, Word2Vec, and BERT for representation. Finally, regression models such as SVM and DNN are used to predict views, evaluated by MAE and MSE.

1 Introduction

These instructions are for researchers exploring how text mining can be used to analyze song lyrics and features to predict the number of views a song receives. The methods applied are not self-contained, and the general principles of text mining and regression models must be followed. This document provides additional details on preprocessing, feature extraction, and the algorithms used for the analysis. The templates include lyrics, title, feature, and artist data, as well as the views count, which serve as the target variable for prediction.

2 Dataset

To analyze the influence of song lyrics and features on views, we will use the Genius Song Lyrics dataset, which is available on Kaggle. This dataset includes fields such as lyrics, title, features, and artist information. Table 1 displays examples of data points. The views of each song will be our target variable. For this project, we will focus on songs in English only, excluding those in other languages.

From the dataset include:

- **Input features:** Lyrics, title, feature, artist

- **Output label:** Views

This dataset is well-suited for our text mining tasks and will be the primary source for our analysis.

3 Methods

3.1 Preprocessing

To prepare the data for text mining, several preprocessing steps are necessary to clean and normalize the song data. These steps ensure that the text and features are suitable for the representation and regression models applied in the next stages.

- **Lemmatization:** We apply lemmatization to reduce each word to its base or dictionary form, ensuring that variations of a word (e.g., "running," "ran") are treated as the same token.
- **Song Part Indicator Removal:** All non-lyrical parts of the text, such as "[Chorus]" and "[Verse]," are removed to focus solely on the meaningful content of the song lyrics.
- **Feature Cleaning:** Punctuation and other irrelevant symbols within the feature column (e.g., artist collaborations or special characters) are removed to prevent them from influencing the text representations.

The cleaned and preprocessed data will then be used in subsequent steps for generating representations and applying predictive models.

3.2 Representation

To predict views, we first need to generate song representations. The process for generating these representations is illustrated in Figure 1. A song document is defined as the concatenation of "features," "title," and "lyrics." This document will be evaluated using four encoding methods: TF-IDF,

Title	Tag	Artist	Year	Views	Features
<i>Killa Cam</i>	rap	Cam'ron	2004	173,166	{"Cam'ron", "Opera Steve"}
<i>Can I Live</i>	rap	JAY-Z	1996	468,624	{}
<i>Forgive Me Father</i>	pop	Fabulous	2003	4,743	{}

Lyrics	ID	Language (cld3)	Language (ft)
[Chorus: Opera Steve & Cam'ron] Killa Cam, Killa Cam, Cam Killa Cam, Killa Cam Killa Cam, Cam Killa ...	1	en	en
[Produced by Irv Gotti] [Intro] Yeah, hah, yeah, Roc-A-Fella We invite you to somethin' epic, you k...	3	en	en
Maybe cause I'm eatin And these bastards fiend for they grub I carry pumps like I serve gasoline to ...	4	en	en

Table 1: Examples of data points

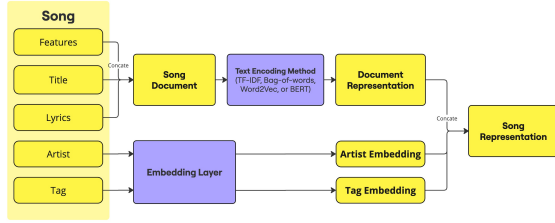


Figure 1: Song representation generation process

Bag-of-Words, Word2Vec (Mikolov et al., 2013), and BERT (Devlin et al., 2019) to generate its representation.

- **TF-IDF:** We use the TF-IDF matrix as the document representation.
- **Bag-of-Words:** We employ both binary and presence indicators for the document representation.
- **Word2Vec:** We use the average word embedding of all tokens.
- **BERT:** We utilize the [CLS] embedding as the document representation.

Other features, such as "artist" and "tags", will be treated as individual nominal attributes. Each nominal attribute will pass through a simple embedding layer to generate its attribute embedding. The final song representation will be the concatenation of the document embedding and all other attribute embeddings.

3.3 Regression

After generating the song representation, we will train a regression model to predict the views. The following regression methods will be applied:

- **SVM:** For song representations where the document representation is generated by either

Model	MAE	MSE
TF-IDF + SVM		
Bag-of-Words + SVM		
Word2Vec + Fully Connected Layer		
BERT + Fully Connected Layer		

Table 2: Example commands for accented characters, to be used in, e.g., BibTeX entries.

TF-IDF or Bag-of-Words, we will use Support Vector Machines as the regression method.

- **Fully Connected Layer:** For song representations where the document representation is derived from Word2Vec or BERT, we will use a fully connected layer for regression. We will try Adam and SGD for optimization.

4 Evaluation

The dataset will be divided into training and testing sets at a ratio of 9 : 1. After training the prediction model, we will evaluate its performance using the methods outlined below. For models with document representations generated by Word2Vec and BERT, these evaluation methods will also be applied to the loss function in the fully connected layer.

- **MAE:** Mean absolute value, measures the average of absolute differences between the predicted views and actual views.
- **MSE:** Mean Squared Error, measures the average of squared differences between the predicted views and actual views.

Table 2 presents a sample evaluation matrix.

References

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [Bert: Pre-training of deep](#)

bidirectional transformers for language understanding.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. [Efficient estimation of word representations in vector space](#).

OpenAI. 2024. [Chatgpt](#). Polished text using ChatGPT.